z-score

A normal curve is determined by mean and SD . If the data follow the normal curve , than knowing mean and SD means knowing the whole histogram.

To compute area under the normal curve , we first standardize the data using Z score formula.

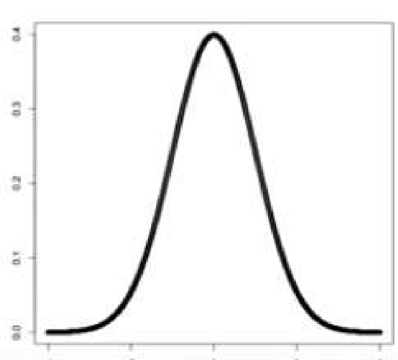
$$Z = \frac{(observed\ value\ - mean)}{SD}$$

Here Z is called the standardized value or Z scaore.

Z has no unit (the observed value, mean,SD have the unit for example meter , inches etc.)

When we standarized the data we have mean 0 and standard deviation equal to 1 →this is the point of stansardization.

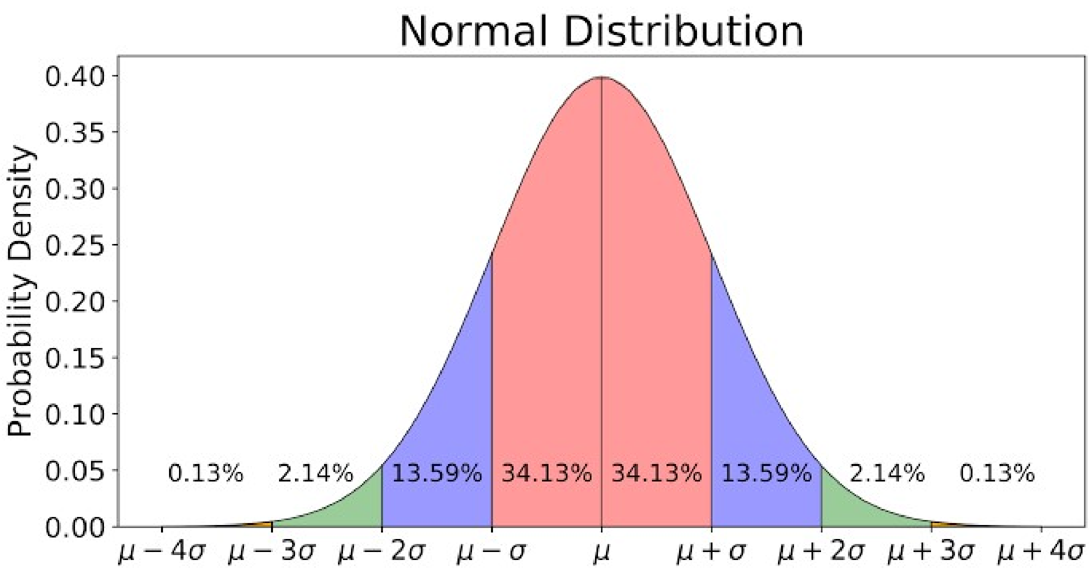
We can convert any normal distribution into the standard normal distribution



This curve is given by the function

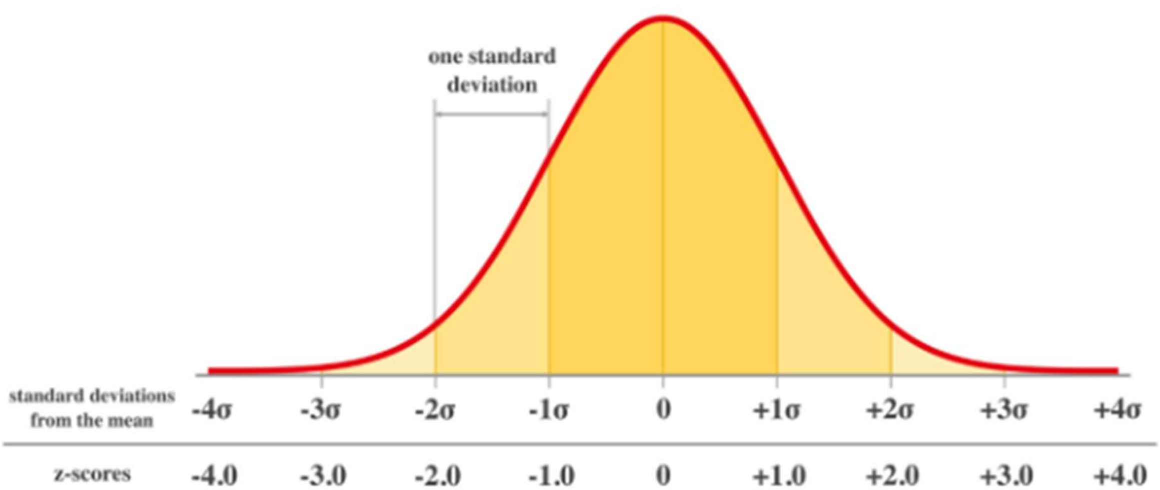
$$y = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

No need to Remember this formula it is just for an understanding.



Standard Normal Distribution

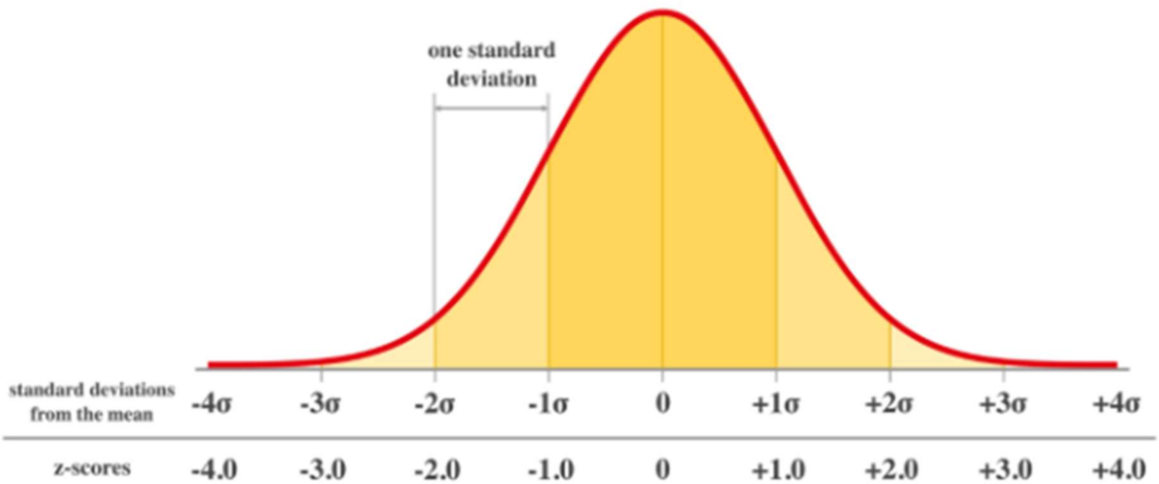
The (z-value/ z-score / z / standard score) represents the number of standard deviations an observation is from the mean for a set of data.



Z-scores range from -3 standard deviations (which would fall to the far left of the normal distribution curve) up to +3 standard deviations (which would fall to the far right of the normal distribution curve). In order to use a z-score, you need to know the mean μ and also the standard deviation σ.

- **Z-scores can be positive or negative.**

- The sign tells you whether the observation is above or below the mean.
- z-score of +2 indicates that the data point two standard deviations above the mean,
- z-score of -2 signifies it is two standard deviations below the mean.



$$Z = \frac{(observed\ value\ - mean)}{SD}$$

we can use the z-value.

```
import scipy.stats as stats

stats.zscore(data=df, axis=1)

or
df.apply(stats.zscore)
```

Normal approximation:-

Finding areas under the normal curve is called normal approximation.

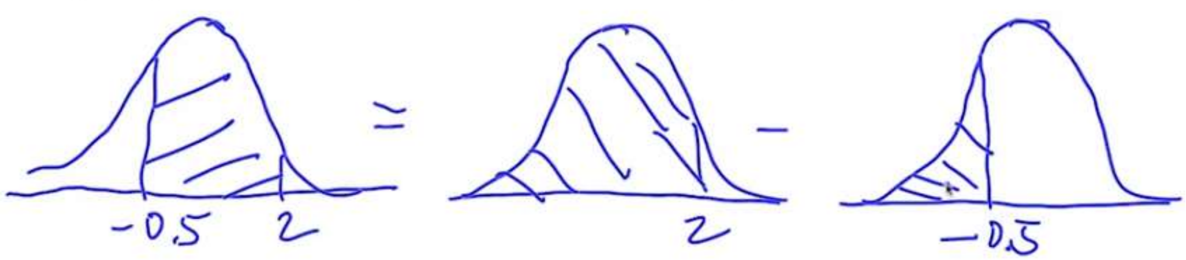
Q. fathers' heights follow the normal curve with a mean of 68.3 inches and a standard deviation of 1.8 inches.
What percentage of fathers have heights between 67.4 inch and 71.9 inch?

1. Standardize:

$$\frac{67.4 \text{ in} - 68.3 \text{ in}}{1.8 \text{ in}} = -0.5 \qquad \frac{71.9 \text{ in} - 68.3 \text{ in}}{1.8 \text{ in}} = 2$$

desired area in a form that can be computed by looked up in a table:

Typically we can look up the area to the left of a given value.



Use a table to find these values: $97.7\% - 30.9\% = 66.8\%$

Q. If you know that -0.1 corresponds to approximately 46 % and 1.8 corresponds to approximately 96.4 % (both percentages are areas under the curve to the left of the value), what percentage of fathers will have heights between 68.1 inc and 71.5 in?

Ans:-

Q. What is the 30 percentile of the Father's height?

Ans:-

What is the need of standarization?

Example :-

One student score 80 marks in Mathematics and 75 Marks in English.

→At this point we can say he has performed excellent in Maths as compare to english.

Consider the average class score of Mathematics is 90, SD is 5.

Average class score of English is 60 , SD is 5.

→lets calculate the Z values

Zm=(80-90)/5=-2

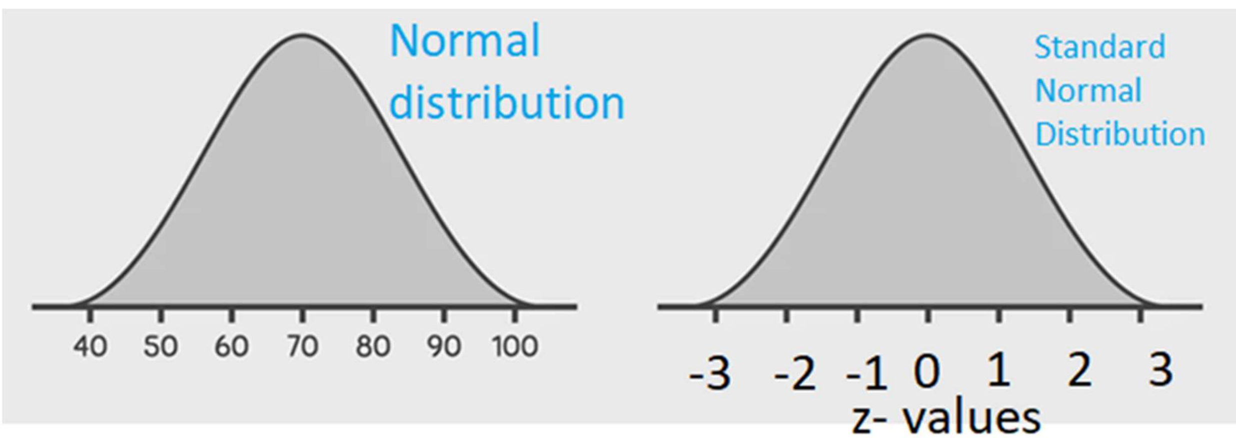
Ze=(75-60)/5=3

Z score value is -3 to +3

if it is -3 and close to it means lower perform and z score close to +3 means excellent performance.

By the use of Z-table we can find the Area under the curve / percentile value.

Conversion of normal distribution to Standard Normal Distribution.



Q. For a recent final written statistics exam for a "Data scientist" job selection process, the mean was 70 with a standard deviation of 10. If you scored 76 marks. What is your percentile or (area in the Normal distribution)?.

Here, mean=70

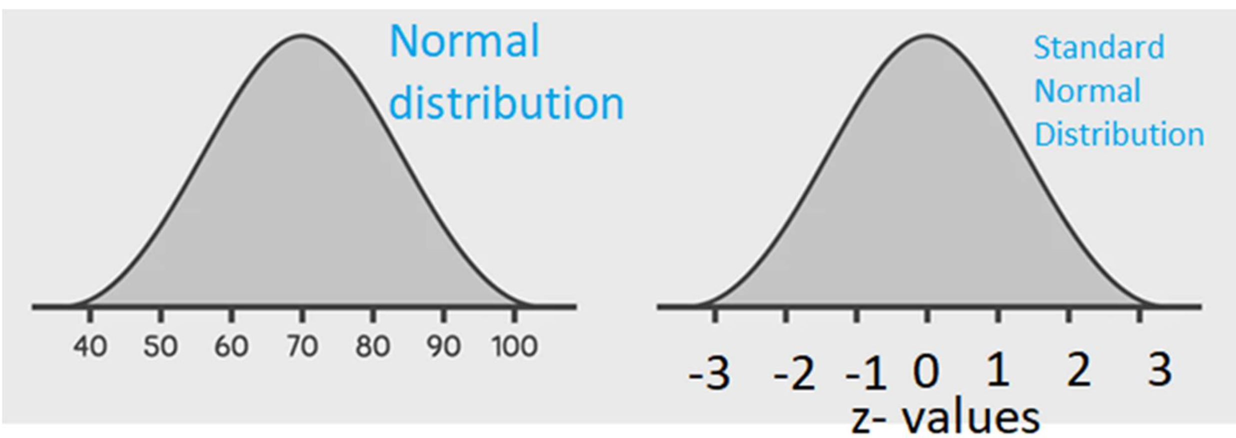
SD=10.

$$Z = \frac{(observed\ value\ - mean)}{SD}$$

Z= (70-70)/10=0
Z=(80-70)/10=1
Z=(60-70)/10=-1
Z=(76-70)/10=0.60

In the z table the value of 0.60 is 0.7257. This is the value of area under curve or the percentile.

Q.1 What proportion of students scored less than 49 % of marks.?



Q. What proportion of students are between 5.81 feet & 6.3 feet height. Given Mean=5.5, sd=0.5 feet.

Q. mean height of Gurkhas is 146 cm with Sd of 3 cm . what is the probability of

(a)Height having greater than 152 cm.

(b)Height between 140 and 150 cm.

Q. Mean demand of an oil is 1000 ltr per month with SD Of 250 ltr.

(a)if 1200 ltrs are stocked , what is the satisfaction level?

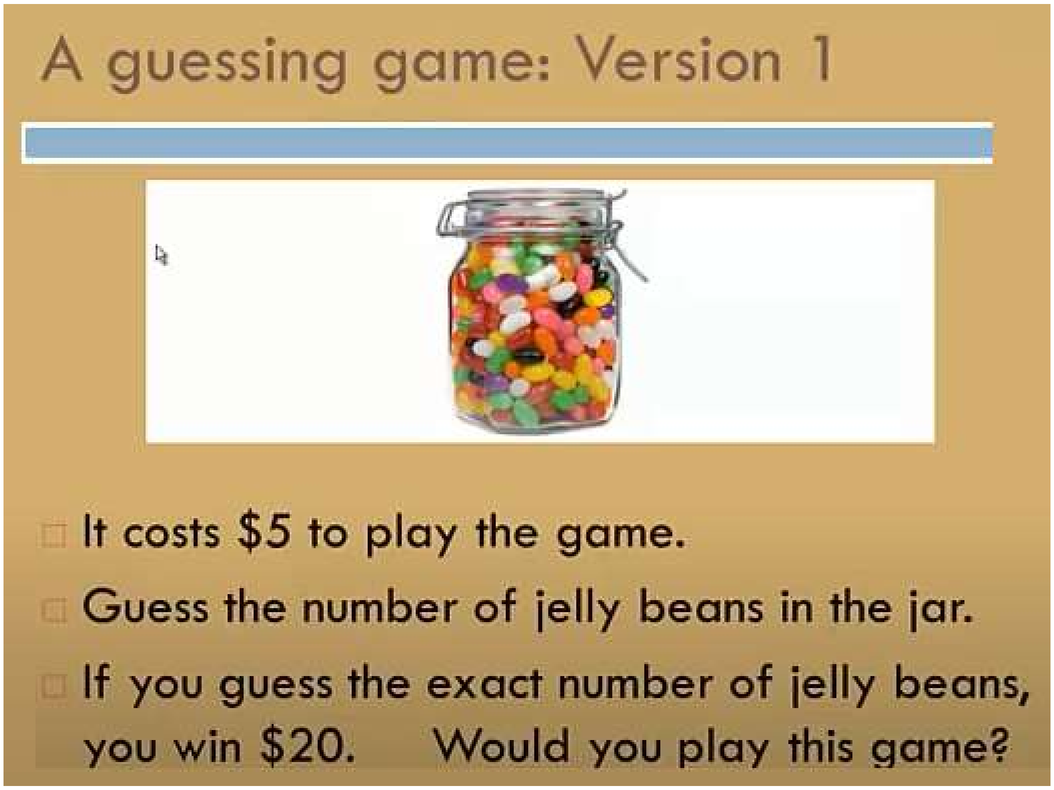
(b)For an assurance of95%.what stock must be kept?

Limitations of Z-Score

Though Z-Score is a highly efficient way of detecting and removing outliers, we cannot use it with every data type. When we said that, we mean that it only works with the data which is completely or close to normally distributed, which in turn stimulates that this method is not for skewed data, either left skew or right skew.

For the other data, we have something known as Inter quartile range (IQR) method.

Confidence Level and Margin of Error

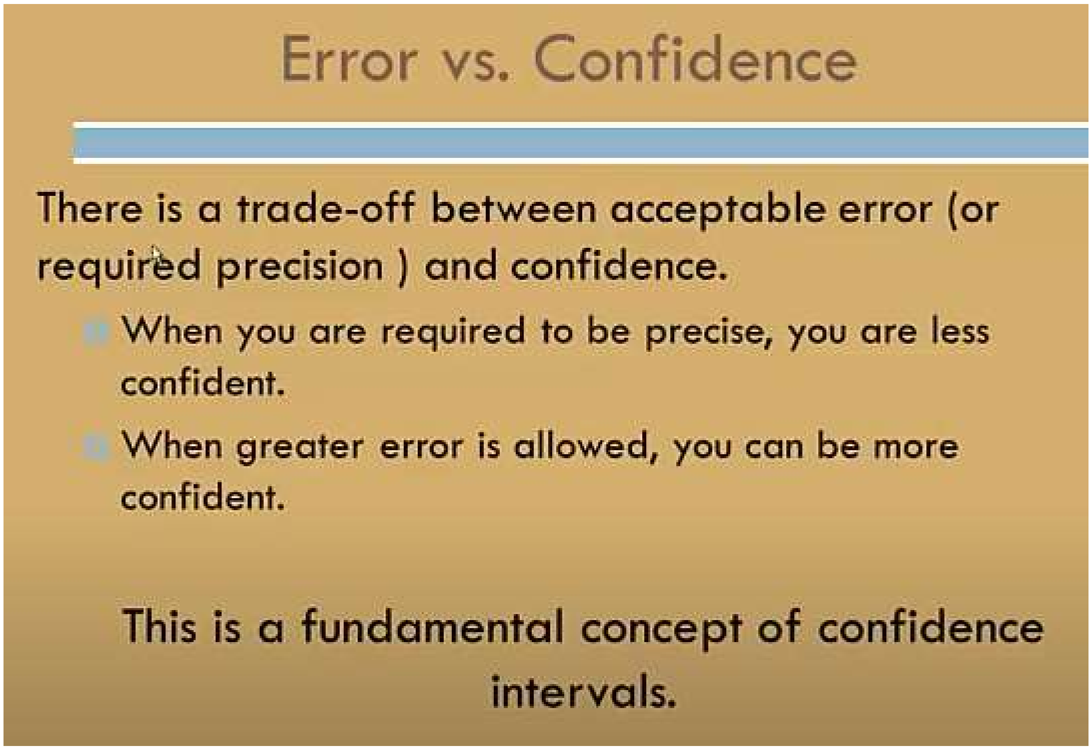


Option 1:- You Guess within 5 no. of actual no. of jelly.

Option 2:- You Guess within 25 no. of actual no. of jelly.

Option 3:- You Guess within 50 no. of actual no. of jelly.

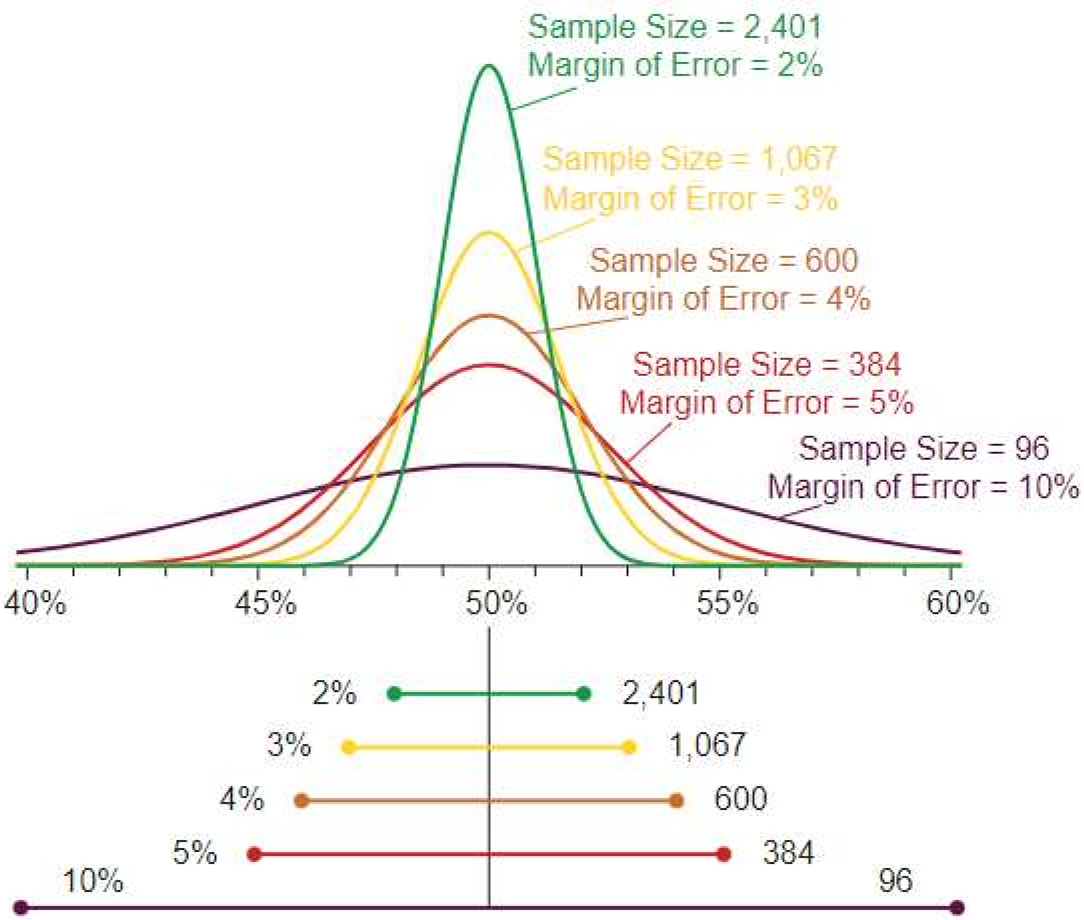
Option 4:- You Guess within 100 no. of actual no. of jelly.



An increase in confidence level results in the increase in the margin of error. We have to choose between the precision and confidence.

Calculating Sample Size

- **To determine a sample size that will provide the most meaningful results, researchers first determine the preferred margin of error (ME) or the maximum amount they want the results to deviate from the statistical mean.**
- **It's usually expressed as a percentage, as in plus or minus 5 percent. Researchers also need a confidence level, which they determine before beginning the study.**
- **This number corresponds to a Z-score, which can be obtained from tables. Common confidence levels are 90 percent, 95 percent and 99 percent, corresponding to Z-scores of 1.645, 1.96 and 2.576 respectively.**
- **Researchers express the expected standard of deviation (SD) in the results. For a new study, it's common to choose 0.5.**



Having determined the margin of error, Z-score and standard deviation, researchers can calculate the ideal sample size by using the following formula:

$$(\text{Z-score})^2 \times \text{SD} \times (1-\text{SD})/\text{ME}^2 = \text{Sample Size}$$

This formula does not depend on the size of the population, only on the size of the sample.

Effects of Small Sample Size

- **In the formula, the sample size is directly proportional to Z-score and inversely proportional to the margin of error.**
- **Consequently, reducing the sample size reduces the confidence level of the study, which is related to the Z-score.**
- **Decreasing the sample size also increases the margin of error.**
- **In short, when researchers are constrained to a small sample size for economic or logistical reasons, they may have to settle for less conclusive results.**
- **Whether or not this is an important issue depends ultimately on the size of the effect they are studying.**